

Practical 11:

```
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define INCREMENTS 100000

int counter = 0;

void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        counter++;
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    printf("Initial counter: %d\n", counter);

    // Create threads
    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment_counter, NULL);
    }

    // Wait for all threads
    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter: %d\n",
```

```
// Wait for all threads
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Expected counter: %d\n",
      NUM_THREADS * INCREMENTS);

printf("Actual counter: %d\n", counter);

return 0;
}
```

```
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define INCREMENTS 100000

int counter = 0;

pthread_mutex_t lock;

void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    // Initialize mutex
    pthread_mutex_init(&lock, NULL);

    printf("Initial counter: %d\n", counter);

    // Create threads
    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL,
```

```

// Create threads
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_create(&threads[i], NULL,
                  increment_counter, NULL);
}

// Wait for threads
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Expected counter: %d\n",
       NUM_THREADS * INCREMENTS);

printf("Actual counter: %d\n", counter);

// Destroy mutex
pthread_mutex_destroy(&lock);

return 0;
}

```

```

bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p pthread_lab
cd pthread_lab
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ nano race_condition.c
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ gcc -Wall -Wextra -pthread -o race_condition race_condition.c
race_condition.c: In function 'increment_counter':
race_condition.c:9:31: warning: unused parameter 'arg' [-Wunused-parameter]
   9 | void *increment_counter(void *arg)
     |                             ^~~~~~
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ ./race_condition
Initial counter: 0
Expected counter: 400000
Actual counter: 115856
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ nano mutex_counter.c
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ ^C
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ gcc -Wall -Wextra -pthread -o mutex_counter mutex_counter.c
mutex_counter.c: In function 'increment_counter':
mutex_counter.c:11:31: warning: unused parameter 'arg' [-Wunused-parameter]
   11 | void *increment_counter(void *arg)
     |                             ^~~~~~
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ ./mutex_counter
Initial counter: 0
Expected counter: 400000
Actual counter: 400000
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ pthread_mutex_destroy(&lock);
-bash: syntax error near unexpected token `&'
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ pthread_mutex_destroy(&lock);
-bash: syntax error near unexpected token `&'
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ cd ~/pthread_lab
nano race_condition.c
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ gcc -Wall -Wextra -pthread -o race_condition race_condition.c
./race_condition
race_condition.c: In function 'increment_counter':
race_condition.c:9:31: warning: unused parameter 'arg' [-Wunused-parameter]
   9 | void *increment_counter(void *arg)
     |                             ^~~~~~
Initial counter: 0
Expected counter: 400000
Actual counter: 278906
bhanu@DESKTOP-DE32HQ8:~/pthread_lab$ |

```